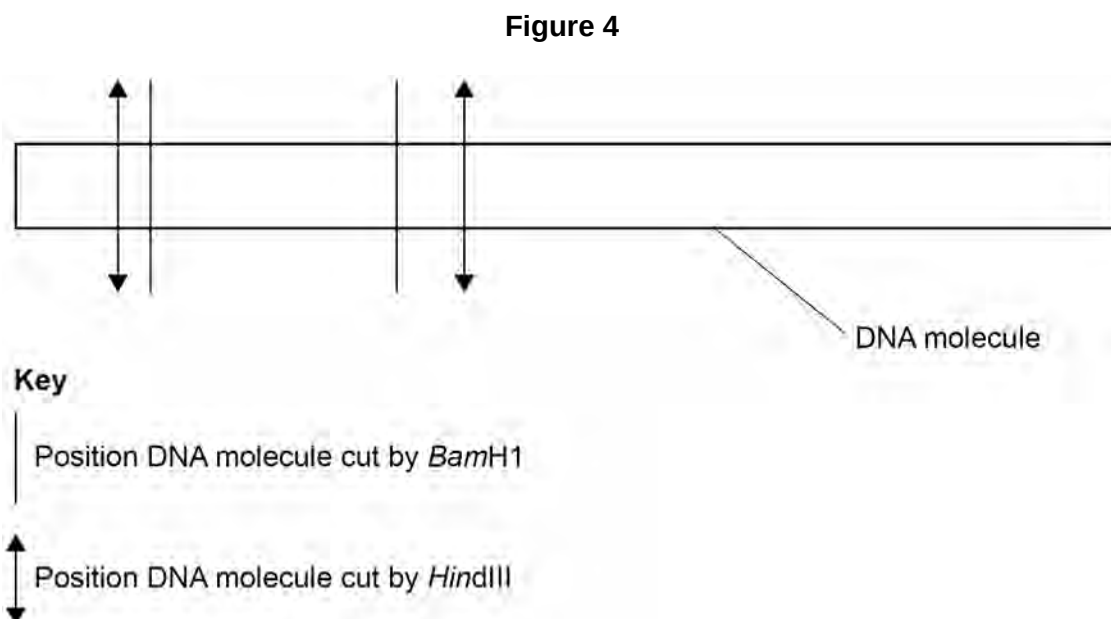


07.1 *Bam*H1 and *Hind*III are both restriction endonucleases.

Figure 4 shows the positions where these enzymes cut a linear molecule of DNA.



In two experiments, multiple copies of the DNA molecule shown in **Figure 4** were all completely cut into fragments using these restriction enzymes. The DNA fragments produced were then separated by electrophoresis.

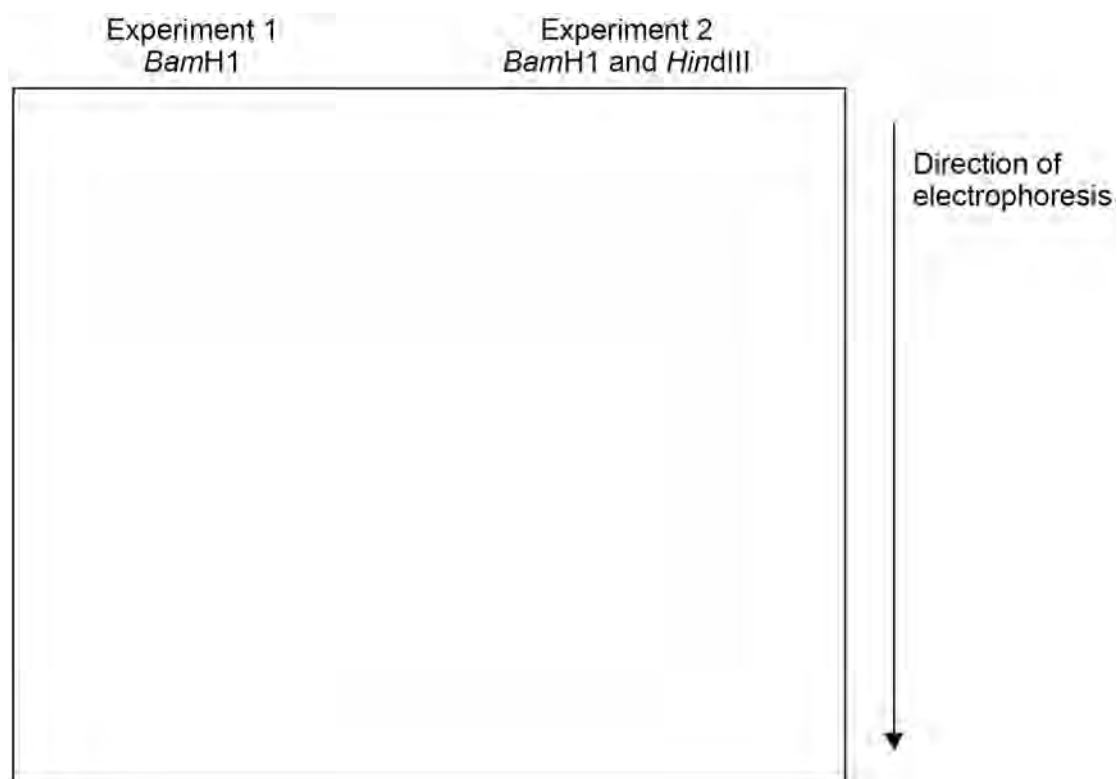
Experiment 1 – DNA cut into fragments using *Bam*H1.

Experiment 2 – DNA cut into fragments using *Bam*H1 and *Hind*III.

Complete **Figure 5** to show the relative positions of the bands following electrophoresis in experiments 1 and 2.

[2 marks]

Figure 5



0 7 . 2

Suggest how you could determine the size of the different DNA fragments produced in these experiments.

[2 marks]

0 7 . 3

How many DNA fragments would be produced in experiment 2 if the original DNA molecule was a plasmid?

[1 mark]

0 7 . 4

Describe how restriction endonuclease and DNA ligase are used to insert a gene into a plasmid.

[2 marks]

E

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